THE CHOW RANK OF THE FIVE-BY-FIVE PERMANENT

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Abstract. We prove that the ordinary Chow rank of the permanent of a generic five by five matrix is sixteen over every field of characteristic zero. The lower bound uses an image-span inequality for rectangular catalecticants and a quotient-symbol estimate for a single quintic product. The finite verification records a 352-case squarefree table, four small rational derivative matrices, and elementary partition bounds. Their inputs and independent verification programs are included. The proof makes no assertion about border Chow rank.

1. Statement and derivative spaces

Let $\mathbb{K}$ initially be algebraically closed of characteristic zero, and put

$$V = \operatorname{span}_{\mathbb{K}}\{x_{ij} : 1 \le i, j \le 5\}, \qquad f = \operatorname{perm}_5 = \sum_{\sigma \in S_5} \prod_{i=1}^{5} x_{i,\sigma(i)}.$$

Here S_5 is the permutation group on $\{1, \dots, 5\}$. For a degree-m form $g \in \operatorname{Sym}^m V$, define

$$\begin{aligned} C_{a,m-a}(g) &: \operatorname{Sym}^a V^* \longrightarrow \operatorname{Sym}^{m-a} V, \\ C_{a,m-a}(g)(\lambda_1 \cdots \lambda_a) &:= \partial_{\lambda_1} \cdots \partial_{\lambda_a} g, \\ \mathcal{D}_j(g) &:= \operatorname{im} C_{m-j,j}(g). \end{aligned}$$

The variables are members of V, hence linear functions on V^*. For a subspace G of forms, ∂G is the span of all its first directional derivatives. For a quadratic form, rank means the rank of its symmetric bilinear matrix. Its essential-variable space is the least $L \subset V$ supporting it; this has dimension equal to its matrix rank.

The ordinary Chow rank $\operatorname{ChowRank}_{\mathbb{K}}(g)$ is the least number of nonzero products of m linear forms summing to g, with $\operatorname{ChowRank}_{\mathbb{K}}(0) = 0$. It is also called product rank [3]. Han and Song [1] prove rank sixteen for the five by five permanent tensor, regarded as a tensor with five separate row factors. Tensor decompositions require each linear factor to belong to its assigned row space; an ordinary Chow decomposition imposes no such requirement. Thus that tensor-rank lower bound does not by itself give the Chow-rank lower bound proved here.

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Theorem 1.1. *Over every field $\mathbb{K}$ of characteristic zero,*

$$\mathrm{ChowRank}_{\mathbb{K}}(\mathrm{perm}_5) = 16.$$

Put $E_j = \mathcal{D}_j(f)$. We will use the following elementary facts.

Lemma 1.2. *One has $\dim E_2 = \dim E_3 = 100$, $\partial E_2 = V$, and*

$$E_2^{(1)} := \{g \in \mathrm{Sym}^3 V : \partial g \subset E_2\} = E_3.$$

Every nonzero quadratic in E_2 has matrix rank at least four. If $L \subset V$ has dimension at most five, then

$$\dim(E_2 \cap \mathrm{Sym}^2 L) \le 1.$$

Proof. The subpermanents on two rows and two columns form a basis of E_2, and those on three rows and three columns form a basis of E_3. Their matching supports are disjoint, and both counts are $\binom{5}{2}^2 = 100$. Differentiating a rectangle at one corner gives the opposite variable.

If every derivative of a cubic lies in E_2, a monomial with a repeated row or column is impossible: some first derivative would contain a monomial outside the matching support of E_2. Inside one three by three block, the rectangle relations equate the coefficients of matchings differing by a transposition. Transpositions connect the six matchings. Thus the cubic is a linear combination of the three by three subpermanents, proving the prolongation assertion.

The row-column diagonal torus acts with distinct characters on the rectangle subpermanents. A generic one-parameter subgroup takes any nonzero quadratic in E_2, projectively, to a single rectangle. Matrix rank cannot increase under this specialization, and a rectangle has rank four.

For completeness, the coordinate-specialization argument needed for the intersection bound is as follows. Choose a one-parameter subgroup whose weights on the 25 variables are distinct. In the Grassmannian of a-planes, the limit of any L is a coordinate a-plane: the limit is fixed by the subgroup, whose variable eigenspaces are one-dimensional. The dimension of $E_2 \cap \mathrm{Sym}^2 L$ is upper semicontinuous, since it is a kernel dimension of a vector-bundle map. It suffices to bound it at the coordinate limit. A coordinate plane corresponds to a bipartite graph with $a \le 5$ edges, and its intersection dimension counts four-cycles. Two different four-cycles use at least six edges. The bound is therefore one. □

2. A quotient-symbol estimate for quintic products

Let $T = \ell_1 \cdots \ell_5 \neq 0$, and set

$$L = \mathrm{span}\{\ell_1, \dots, \ell_5\}, \qquad F = \mathcal{D}_2(T), \qquad U = \mathcal{D}_3(T),$$
$$u = \dim U = \dim F \le 10.$$

The equality of dimensions follows by transposing the two catalecticants under the perfect differential pairing in characteristic zero. The upper bound follows because third derivatives are spanned by the ten pair products. Let

$R = F \cap E_2$. Lemma 1.2 gives $\dim R \leq 1$, and each nonzero element of R has quadratic rank at least four.

For a surjection $P : L \twoheadrightarrow D$, of rank d, let $q_R : F \to F/R$ be the quotient map. Polarization is

$$\delta : U \longrightarrow L \otimes F, \qquad g \longmapsto \sum_j y_j \otimes \partial_{y_j^*} g,$$

where (y_j) is any basis of L and (y_j^*) its dual. It is basis-independent. Write $\beta = (P \otimes q_R)\delta$.

Lemma 2.1 (Small complementary span)**.** *If $k = \dim \ker P \leq 2$, then*

$$\ker \beta = U \cap \operatorname{Sym}^3(\ker P), \qquad \operatorname{rank} \beta \geq \max\{0, u - \tbinom{k+2}{3}\}.$$

Proof. Choose coordinates consisting of a basis of $S = \ker P$ and d complementary variables. For a kernel cubic, its derivatives in all the complementary directions lie in R. If one is nonzero, changing the complementary coordinates makes these derivatives $q, 0, \ldots, 0$, with $0 \neq q \in R$. Characteristic zero then implies that the cubic, and hence q, uses only S and the first complementary variable. This gives $\operatorname{rank} q \leq k + 1 \leq 3$, a contradiction. All complementary derivatives vanish, which is precisely membership in $\operatorname{Sym}^3 S$. The dimension bound follows. □

Lemma 2.2 (Five independent factors)**.** *If $\dim L = 5$, then $u = 10$ and $\operatorname{rank} \beta \geq 2d$.*

Proof. Use the factors as coordinates $z_1, \ldots, z_5$. Then U and F are the squarefree cubic and quadratic spaces. For arbitrary $(\ker P, R)$, simultaneously specialize in their Grassmannians by a generic one-parameter subgroup of the diagonal torus. Its weights on the variables and on the quadratic monomials can be chosen distinct, so both limits are coordinate subspaces. The rank of the induced map of the tautological quotient bundles cannot increase in this limit. No rank-four condition on the limiting R is required here.

At a coordinate point let B index the kernel of P, with $|B| = 5 - d$, and let A be the empty set or a single killed quadratic edge. Different source triples have disjoint polarization supports. A cubic column is killed exactly when it is contained in B, or has one vertex outside B and its remaining edge is the unique edge of A contained in B. The number of killed columns is therefore

$$\binom{5-d}{3} + d\,\mathbf{1}_{\{A=\{e\},\ e\subset B\}}.$$

Taking the minimum rank over these possibilities gives

d	0	1	2	3	4	5
$\operatorname{rank}\beta$ lower bound	0	5	7	7	10	10.

Each entry is at least $2d$. The entire coordinate domain has $2^5(1+\binom{5}{2}) = 352$ pairs and is also checked directly by the supplied programs. □

Lemma 2.3 (Four-dimensional factor span)**.** *If* $\dim L = 4$*, then* $u \geq 7$*. For* $d = 1$*,*

$$\operatorname{rank}\beta + (10-u)/2 \geq 2.$$

Proof. Choosing four independent factors and rescaling coordinates puts the product in one of the forms

$$T_s = x_1x_2x_3x_4(x_1 + \cdots + x_s), \qquad 1 \leq s \leq 4.$$

Order-two differentiation, over $\mathbb{Q}$, gives

$$\begin{array}{c|cccc} s & 1 & 2 & 3 & 4 \\ \hline u & 7 & 7 & 9 & 10. \end{array} \tag{2.1}$$

These are ranks of the coefficient matrices of the ten second partials in the twenty degree-three monomials on four variables. Appendix A gives explicit nonzero minors and upper-rank relations; the ancillary programs independently check the complete matrices. Since their coefficients are rational, the ranks hold over every characteristic-zero field.

For $s = 1, 2$, the actual cubic derivative space contains $x_1x_3x_4$ and $x_2x_3x_4$. Every nonzero direction differentiates at least one of these to a nonzero quadric on at most three variables. That quadric has rank at most three and hence cannot belong to R. Thus $\operatorname{rank}\beta \geq 1$, which with $u = 7$ is more than enough.

For $s = 3$, the actual space contains $x_1x_2x_4, x_1x_3x_4, x_2x_3x_4$. Write the direction as $\lambda = \sum_{j=1}^{4} a_j\partial_{x_j}$. If $a_4 \neq 0$, projecting their derivatives modulo x_4L gives three independent pair monomials, so quotienting by R leaves rank at least two. If $a_4 = 0$, their derivatives span a space of dimension at least two divisible by x_4: if, for example, $a_1 \neq 0$, the first two have independent linear factors. Every quadric in this space has rank at most two, so its intersection with R is zero. Again $\operatorname{rank}\beta \geq 2$.

For $s = 4$, the actual space contains all four squarefree cubics. Choose $a_j \neq 0$; differentiating the three cubics containing x_j, modulo x_jL, gives three independent pair monomials. The quotient by R has rank at least two. The stated inequality follows for $u = 9, 10$. All the cubic-space inclusions used here are checked directly along with (2.1), rather than inferred from formal product spans. □

Proposition 2.4 (Half-defect estimate)**.** *For every nonzero quintic Chow term and every quotient of its actual factor span,*

$$\boxed{\operatorname{rank}\beta + \frac{10-u}{2} \geq 2d.}$$

Proof. The independent case is Lemma 2.2. For factor-span dimension $\ell \leq 4$, the middle rank is bounded above by $\min\{10, \binom{\ell+1}{2}\}$, because $u = \dim F$. It is bounded below by $1, 2, 4, 7$, respectively for $\ell = 1, 2, 3, 4$. Indeed, choose ℓ independent factors as coordinates and specialize every remaining factor to a coordinate initial by a generic diagonal one-parameter subgroup. The

resulting monomial uses all ℓ variables and has total degree five. Counting its degree-two divisors, over the positive partitions of five, gives these minima. Catalecticant rank cannot increase under specialization.

If $\ell \le 3$, every $d > 0$ has complementary span at most two. Lemma 2.1, minimized over the indicated integer interval for u, gives the following lower bounds for $\operatorname{rank}\beta + (10-u)/2$:

ℓ	$d = 0, \dots, \ell$
1	$(9/2, 11/2)$
2	$(7/2, 5, 6)$
3	$(2, 3, 6, 7)$.

For $\ell = 4$, the same lemma covers $d = 2, 3, 4$ and gives $9/2, 15/2, 17/2$, respectively. The case $d = 0$ is immediate, and $d = 1$ is Lemma 2.3. Every bound is at least $2d$. □

3. Two global estimates

Lemma 3.1 (Rectangular image-span inequality)**.** *Let $A_i : X^* \to Y$ be linear maps of ranks u_i. If $D_Y = \dim \sum_i \operatorname{im} A_i$ and $D_X = \dim \sum_i \operatorname{im} A_i^*$, then*

$$\operatorname{rank}\Big(\sum_i A_i\Big) \ge D_Y + D_X - \sum_i u_i.$$

Proof. Factor $A_i = B_i C_i$, with $B_i : \mathbb{K}^{u_i} \to Y$ injective and $C_i : X^* \to \mathbb{K}^{u_i}$ surjective. The horizontal concatenation B has rank D_Y, and the vertical concatenation C has rank D_X, since its dual image is $\sum_i \operatorname{im} A_i^*$. Now $\sum_i A_i = BC$; Sylvester's rank inequality proves the claim. □

Suppose $f = \sum_{i=1}^N T_i$, with nonzero quintic Chow terms. Put

$$F_i = \mathcal{D}_2(T_i), \quad U_i = \mathcal{D}_3(T_i), \quad u_i = \dim F_i = \dim U_i, \quad \Delta = \sum_i (10 - u_i),$$

$$F = \sum_i F_i, \quad U = \sum_i U_i, \quad \dim F = 100 + h_2, \quad \dim U = 100 + h_3.$$

Linearity of the catalecticants gives $E_2 \subset F$ and $E_3 \subset U$, so $h_2, h_3 \ge 0$. Under the differential pairings, the transpose of $C_{3,2}(T_i)$ is $C_{2,3}(T_i)$. Lemma 3.1, applied to their sum, gives

$$h_2 + h_3 \le 10N - 100 - \Delta. \tag{3.1}$$

Let $q : F \to F/E_2$. The map

$$\widetilde{\beta} : \bigoplus_i U_i \longrightarrow V \otimes (F/E_2), \qquad (g_i) \longmapsto \sum_i (1_V \otimes q)\delta(g_i)$$

factors through addition into U. By Lemma 1.2, its kernel after addition is exactly E_3, hence $\operatorname{rank}\widetilde{\beta} = h_3$.

Set $L_i = \operatorname{span}\{\text{factors of } T_i\}$, choose any order, and put $W_i = L_1 + \cdots + L_i$, $W_0 = 0$, and $d_i = \dim(W_i/W_{i-1})$. These spaces span V, since $V = \partial E_2 \subset$

$\partial F \subset \sum_i L_i$, so $\sum_i d_i = 25$. Let Z_i be the image of the i-th summand of $\widetilde{\beta}$, and $H_i = Z_1 + \cdots + Z_i$. Projection to $(W_i/W_{i-1}) \otimes (F/E_2)$ kills H_{i-1}. Thus

$$\dim H_i - \dim H_{i-1} \geq \operatorname{rank}\big((P_i \otimes q|_{F_i})\delta|_{U_i}\big),$$

where $P_i : L_i \to L_i/(L_i \cap W_{i-1})$ has rank d_i. The injection $F_i/(F_i \cap E_2) \hookrightarrow F/E_2$ shows that this is exactly the local symbol from Proposition 2.4. Summing gives

$$h_3 \geq \sum_i \left(2d_i - \frac{10 - u_i}{2}\right) = 50 - \frac{\Delta}{2}. \tag{3.2}$$

Combining (3.1) and (3.2) yields

$$h_2 + \frac{\Delta}{2} \leq 10N - 150.$$

Consequently $N \geq 15$. If $N = 15$, both h_2 and Δ are zero. Then $F = E_2$, while each F_i has dimension ten and is contained in $E_2 \cap \operatorname{Sym}^2 L_i$, contradicting the one-dimensional bound of Lemma 1.2. Hence $N \geq 16$.

4. Upper bound, scope, and exact verification

Glynn's identity [2] is

$$\operatorname{perm}_5 = \frac{1}{16} \sum_{\substack{\epsilon \in \{\pm 1\}^5 \\ \epsilon_1 = 1}} \left(\prod_{r=1}^{5} \epsilon_r\right) \prod_{c=1}^{5} \left(\sum_{r=1}^{5} \epsilon_r x_{rc}\right).$$

To verify it, expand by row choices in each column. Summation over the four free signs kills a monomial unless each of rows two through five occurs an odd number of times; total degree five then forces every row to occur once. Its coefficient is sixteen before division. This gives a sixteen-term decomposition. Over an arbitrary characteristic-zero field, any hypothetical shorter decomposition remains one after extension to its algebraic closure, where the lower bound applies. The displayed upper bound is rational. This proves Theorem 1.1 over every characteristic-zero field.

The theorem concerns ordinary rank, and proves no border-rank equality. The supplied standard-library verifiers reconstruct the 352 coordinate symbols, the four actual derivative matrices and required cubic inclusions, the positive-partition floors, and every rational inequality used in Proposition 2.4. They also expand the upper bound over all $16 \cdot 5^5 = 50{,}000$ row choices. From the extracted submission root run

```
python -B anc/verify_submission.py
```

The independent finite implementations do not replace the written specialization, rank-four kernel, image-span, or filtration arguments. The earlier September proof package is preserved, with its unchanged manifest, under `anc/original/`. The present lower bound uses the short argument above and does not depend on that package's state routing, terminal orbit classification, or historical private replay receipts.

Appendix A. Exact normal-form minors

This appendix specifies the small calculation in (2.1) without relying on a stored rank value. Let M_s have columns $\partial_{x_i}\partial_{x_j}T_s$, in the order

$$(11, 12, 13, 14, 22, 23, 24, 33, 34, 44).$$

Its rows are indexed by the twenty exponent vectors (e_1, e_2, e_3, e_4) of total degree three in ascending lexicographic order, starting with $(0, 0, 0, 3)$. Indices in the following table start at one. Each indicated square submatrix has the displayed determinant.

s	column indices	row indices	determinant
1	1,2,3,4,6,7,9	6,12,14,15,17,18,19	16
2	1,2,3,4,6,7,9	6,8,9,12,14,15,16	16
3	1,2,3,4,5,6,7,8,9	3,6,7,8,12,13,14,15,17	−16
4	1,2,3,4,5,6,7,8,9,10	2,5,6,7,11,12,13,14,15,16	16

For $s = 1$, columns 5, 8, and 10 are zero. For $s = 2$, columns 8 and 10 are zero, and column 1 minus column 2 plus column 5 is zero. For $s = 3$, column 10 is zero. These give the matching upper ranks $7, 7, 9$; the upper rank ten for $s = 4$ is immediate from the number of columns. Together with the minors this proves (2.1).

The required cubic inclusions have an equally direct check. For $j \le s$,

$$\partial_{x_j}^2 T_s = 2\prod_{i\neq j} x_i.$$

For $s = 1$ this yields $x_2x_3x_4$, and $\partial_{x_1}\partial_{x_2}T_1 = 2x_1x_3x_4$. For $s = 2, 3, 4$, the displayed diagonal derivatives give precisely the squarefree cubics used in Lemma 2.3.

References

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